Subtractive Unlearning for Recommenders*

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Abstract

Compliance with regulations such as GDPR has made machine unlearning a requirement for modern recommender systems. Sequential Recommender Systems (SRSs) are especially challenging, since they rely on temporal interaction patterns: existing methods either require prohibitive full retraining or ignore the sequential nature of user behaviour. We propose SURF (**S**ubtractive **U**pdates for **R**ecommender **F**orgetting), a lightweight, model-agnostic framework for approximate unlearning in SRSs. SURF identifies the embedding neighbourhood of the item to forget, trains an auxiliary model on this compact local subset, and subtracts its scores from those of the original model at inference time. We prove that the mechanism preserves ranking stability, perturbs the scoring function only locally, and removes the influence of the target item up to the approximation error of the auxiliary model. Against five baselines on seven datasets and three backbones, SURF matches full retraining while improving NDCG@20 by up to 32% and using as little as 2% of its time budget. Code: https://github.com/FilippoBetello/SURF.

Keywords

Recommender Systems, Sequential Recommendation, Machine Unlearning, Approximate Unlearning

1. Introduction

Recommender systems shape user experience across e-commerce, entertainment and social media [2, 3, 4], and Sequential Recommender Systems (SRSs) do so by exploiting the temporal structure of user interactions [5, 6]. Data protection laws such as the GDPR [7] and its counterparts worldwide [8] grant a "right to be forgotten", which has motivated machine unlearning: removing the influence of specific data points from a trained model without retraining it from scratch [9]. Early work targeted supervised classifiers [10], and unlearning has since been studied for matrix factorisation and GNN-based recommenders [11], but SRSs remain distinctive: predictions depend on entire interaction histories, so deleting a single item perturbs many learned representations [12].

Existing recommendation unlearning methods fall into two families. Partition-based methods [13, 14, 15] split the training data into shards and retrain only the affected ones, an idea that is exact only for specific model classes such as kNN [16]; in recommendation, however, users and items are densely interconnected, so a target item appears in most shards, which inflates retraining cost and degrades accuracy. Influence-function methods [17, 18] estimate the effect of removal analytically, but rely on assumptions that often break in practice [19] and on expensive second-order computations.

We propose **SURF** (**S**ubtractive **U**pdates for **R**ecommender **F**orgetting), which bridges the two families and requires neither sharding nor second-order approximations. Given an item to forget, SURF identifies the local neighbourhood of its embedding, trains a lightweight auxiliary model on that region only, and subtracts the auxiliary scores from the original ones at inference time (Fig. 1). To the best of our knowledge, this is the first approach introducing model subtraction as an unlearning mechanism in recommender systems. Unlike post-hoc filtering, which leaves internal representations untouched, SURF acts at the representation level, which matters in sequential recommendation, where item removal

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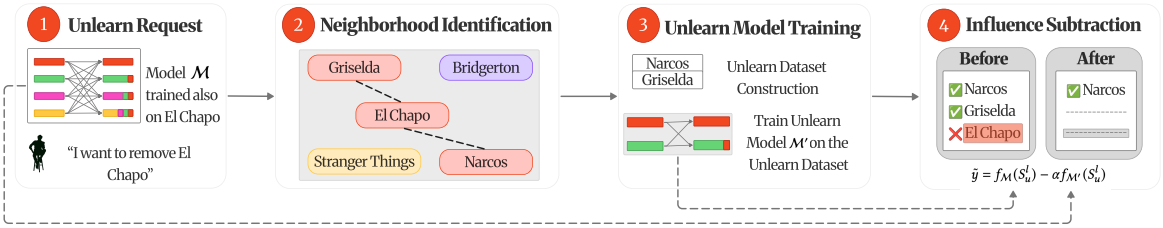


Figure 1: Overview of SURF: given an unlearning request, we identify the neighbourhood of the item to forget, train an auxiliary model on this subset, and subtract its influence at inference time.

has cascading effects along user sequences. Since the original parameters are never modified, forgetting is also instantly reversible, an attractive property in production. SURF is model-agnostic and, while it relies on an approximate but theoretically grounded isolation of item influence, it improves NDCG@20 by up to 32% over existing methods while requiring just 2% of the retraining time budget.

2. Method

Let $\mathcal{U}$ and $\mathcal{I}$ be sets of users and items. Each user u is represented by a sequence of ℓ_u interactions $S_u = (i_1, \ldots, i_{\ell_u})$, $i_j \in \mathcal{I}$. An SRS $\mathcal{M}$ takes the prefix $S_u^l = [i_1, \ldots, i_l]$, returns scores $f_{\mathcal{M}}(S_u^l) \in \mathbb{R}^{|\mathcal{I}|}$ and predicts the next item. Given $\mathcal{M}$ trained on $\mathcal{D}$, an unlearning request specifies an item i^* to forget, with forget set $\mathcal{D}_f = \{S_u : u \in \mathcal{U} \wedge i^* \in S_u\}$; the goal is a model behaving as if it had never been trained on $\mathcal{D}_f$, without retraining from scratch. SURF proceeds in four steps.

1. Neighbourhood identification. Let $\mathbf{e}_{i^*} \in \mathbb{R}^d$ be the embedding of the item to forget and $\mathcal{E} = \{\mathbf{e}_j \mid j \in \mathcal{I}\}$ the set of item embeddings. We retrieve its k nearest neighbours $\mathcal{N}_k(\mathbf{e}_{i^*}) = k\text{-arg min}_{\mathbf{e}\in\mathcal{E}} \|\mathbf{e} - \mathbf{e}_{i^*}\|$, where k-arg min returns the k closest elements. For architectures without user embeddings (e.g. SASRec [20]) we represent users as $\mathbf{e}_u = \frac{1}{\ell_u}\sum_{j=1}^{\ell_u} \mathbf{e}_{i_j}$. We use the pure nearest neighbours of $\mathbf{e}_{i^*}$ rather than only the embeddings of sequences in $\mathcal{D}_f$, since the latter do not fully characterise the local region of the representation space where information about the item is encoded; the neighbourhood better approximates this local manifold, enabling more effective and robust forgetting [21].

2. Unlearn dataset construction. We build a small unlearn dataset from the sequences touching that neighbourhood, $\mathcal{D}_r = \{S_u : \exists\, i \in S_u \mid i \in \mathcal{N}_k(\mathbf{e}_{i^*})\}$, selecting the embeddings most affected by the forgotten item.

3. Unlearn model training. We train a small unlearn model $\mathcal{M}'$ on $\mathcal{D}_r$ with the same architecture and objective as $\mathcal{M}$, so that it approximates the influence of $\mathcal{D}_r$ on predictions.

4. Influence subtraction. At inference time the new scores are:

$$\tilde{y} = f_{\mathcal{M}}(S_u^l) - \alpha f_{\mathcal{M}'}(S_u^l), \qquad \tilde{i}_{l+1} = \arg\max_{i\in\mathcal{I}} \tilde{y}_i, \tag{1}$$

where $\alpha > 0$ controls the unlearning strength.

2.1. Theoretical Analysis

Ranking stability. Let $\Delta_{i_1 i_2} = f_{\mathcal{M}}(S_u^l)_{i_1} - f_{\mathcal{M}}(S_u^l)_{i_2}$. If $\Delta_{i_1 i_2} > 2\alpha \|f_{\mathcal{M}'}(S_u^l)\|_\infty$, the relative ranking of i_1 and i_2 is preserved: by the triangle inequality $\tilde{\Delta}_{i_1 i_2} = \Delta_{i_1 i_2} - \alpha(f_{\mathcal{M}'}(S_u^l)_{i_1} - f_{\mathcal{M}'}(S_u^l)_{i_2}) \geq \Delta_{i_1 i_2} - 2\alpha\|f_{\mathcal{M}'}(S_u^l)\|_\infty > 0$. Items with a sufficiently large margin therefore keep their order, and the forgetting operation mainly affects predictions tied to the target item.

Local perturbation. Assume $f_{\mathcal{M}'}$ is L-Lipschitz in the embedding space, so that nearby embeddings give similar auxiliary predictions, and that the influence of the forgotten item is bounded, $\|f_{\mathcal{M}'}(\mathbf{e}_{i^*})\| \leq C$, which prevents excessively large corrections. For every sequence with $\mathbf{e}_u \in \mathcal{N}_k(\mathbf{e}_{i^*})$,

$$\|\tilde{y} - f_{\mathcal{M}}(S_u^l)\| = \alpha\|f_{\mathcal{M}'}(S_u^l)\| \leq \alpha\big(L\,\mathrm{diam}(\mathcal{N}_k(\mathbf{e}_{i^*})) + C\big), \tag{2}$$

obtained by adding and subtracting $f_{\mathcal{M}'}(\mathbf{e}_{i^*})$ and applying the triangle inequality and the Lipschitz bound. When the neighbourhood is compact and α is controlled, SURF thus modifies the prediction function only locally, preserving global recommendation behaviour.

Influence removal. Assume $f_{\mathcal{M}}(S_u^l) = f_{\text{base}}(S_u^l) + f_{i^*}(S_u^l)$, where f_{i^*} is the contribution of the item to forget, and that the auxiliary model approximates it, $|f_{\mathcal{M}'}(S_u^l) - f_{i^*}(S_u^l)| \leq \varepsilon$. Writing the residual as $(1-\alpha)f_{i^*}(S_u^l) + \alpha(f_{i^*}(S_u^l) - f_{\mathcal{M}'}(S_u^l))$ and applying the triangle inequality gives $|f_{i^*}(S_u^l) - \alpha f_{\mathcal{M}'}(S_u^l)| \leq |1-\alpha|\,|f_{i^*}(S_u^l)| + \alpha\varepsilon$, which for $\alpha = 1$ reduces to ε: the contribution of the target item is removed up to the approximation error of the auxiliary model.

3. Experimental Settings

Datasets and metrics. We use seven public datasets: ML-1M and ML-100k (MovieLens [22]), the Beauty subset of Amazon Reviews [23], Steam, the New York City (FS-NYC) and Tokyo (FS-TKY) subsets of Foursquare [24], and Yelp [25]. They deliberately vary in their suitability for SR [26], and most of them are used in prior unlearning work [14, 11]. Recommendation quality is measured with NDCG [27] and with the Finite Rank-Biased Overlap (FRBO) [28], which quantifies the agreement between the rankings before and after unlearning (↑ is better for both). Forgetting is measured with UnlearnRecall and UnlearnMRR [29]: the former counts how many items requested to be unlearned still appear in the Top-N list, the latter how early they appear (↓ is better). All metrics are computed @20.

Backbones and baselines. We implement SURF on three SRSs: GRU4Rec [30, 31], SASRec [20] and BERT4Rec [32]. Every baseline is trained with the same backbones: **Retrain** (full retraining), **RecEraser** [13] and **UltraRE** [14], which shard the data and aggregate the resulting models, **SRU** [15], which shards by embedding similarity, and **IFRU** [17], which unlearns through a dedicated influence-based loss. Post-hoc filtering is excluded as a competitor: letting $g_{\mathcal{M}}(S_u^l)$ be the last hidden state before the output layer, the distance $\|g_{\mathcal{M}}(S_u^l) - g_{\mathcal{M}^U}(S_u^l)\|_2$ between the original and unlearned models is zero for every sequence under filtering, confirming that the influence of the forgotten item is encoded in the representations, whereas SURF yields non-zero distances.

Implementation. We use EasyRec [33], treat ratings as implicit, and drop users and items with fewer than five interactions [20, 32]. Negatives are sampled from items the user has not consumed [34]. The last interaction of each user is used for testing and the second to last for validation. The sequence length is 200 for MovieLens and Foursquare and 50 otherwise [35]. We average over five target items; the batch size is 128 and all models are trained for 500 epochs with Adam, plus 10 aggregation epochs for methods that need them. After a grid search we set $\alpha = 0.7$ and $k = 1$.

4. Results

Effectiveness. Across all backbones and datasets (Table 1), SURF achieves the best or second-best unlearning metrics (UL MRR and UL Recall). On SASRec it attains the lowest unlearning scores on Beauty and ML-1M, outperforming full retraining in several cases, indicating a more complete removal of the target signal; the same holds for GRU4Rec and BERT4Rec against RecEraser, SRU and UltraRE. This does not come at the expense of utility: SURF reaches near-retrain NDCG on Beauty and surpasses retraining on ML-1M, where with SASRec it improves NDCG@20 by 32% over the strongest baseline, and its FRBO values confirm that ranking stability is maintained while the target influence is removed.

Efficiency. Whereas retraining requires full optimisation, SURF roughly halves training time on ML-1M and is consistently cheaper elsewhere, down to $0.04\times$ on the two largest datasets; shard-based methods instead cost up to $17\times$ retraining while also underperforming in either forgetting or recommendation quality. IFRU is the cheapest to train, but it severely degrades recommendation performance, often causing NDCG to collapse. SURF is also the fastest at inference in most settings, and the gap widens with scale: on Yelp competing methods need from $1.18\times$ to $4.89\times$ the inference time of a retrained model, against $0.11\times$ for SURF with SASRec and GRU4Rec. IFRU, despite its fast training, consistently inflates inference time.

Table 1
Unlearning performance (all @20) and relative training time (TT) w.r.t. full retraining. **Bold** denotes the best model for a dataset, underlined the second best. † indicates a statistically significant result w.r.t. the second-best approach based on a Wilcoxon test with **p**-value **< 0.05**. R stands for Recall.

Dataset	Method	SASRec					GRU4Rec					BERT4Rec				
		NDCG (↑)	UL MRR (↓)	UL R (↓)	FRBO (↑)	TT (↓)	NDCG (↑)	UL MRR (↓)	UL R (↓)	FRBO (↑)	TT (↓)	NDCG (↑)	UL MRR (↓)	UL R (↓)	FRBO (↑)	TT (↓)
Beauty	Retrain	**0.6484**†	0.0016	0.0147	**0.7135**†	1.00×	**0.6330**†	**0.0022**	0.0164	**0.6462**†	1.00×	0.6117	0.0035	**0.0191**	**0.5495**†	1.00×
	RecEraser	0.5268	0.0024	0.0182	0.3311	3.38×	0.4150	0.0025	0.0156	0.2303	3.52×	0.3566	0.0020	0.0200	0.2417	2.77×
	SRU	0.3908	0.0036	0.0182	0.2936	6.90×	0.3690	0.0049	0.0217	0.2131	5.72×	0.3153	0.0047	0.0251	0.2960	54.49×
	UltraRE	0.4224	0.0016	0.0147	0.1911	8.70×	0.1394	0.0025	0.0156	0.0932	13.28×	0.1989	**0.0013**	**0.0122**	0.0790	13.29×
	IFRU	0.3167	0.0047	0.0156	0.0956	**0.10**×	0.0019	0.0036	0.0217	0.0087	**0.07**×	0.0019	0.0028	0.0200	0.0065	**0.12**×
	SURF	0.5640	**0.0015**†	**0.0130**†	0.5128	0.89×	0.5492	0.0024	**0.0148**†	0.5669	0.90×	**0.6251**†	0.0028	0.0165	0.5003	0.84×
ML-1M	Retrain	0.1208	0.0018	0.0076	0.3036	1.00×	**0.1541**†	0.0015	0.0058	0.5251	1.00×	**0.0805**†	0.0013	0.0069	**0.6147**†	1.00×
	RecEraser	0.0417	0.0020	0.0084	0.0771	10.21×	0.0459	0.0013	0.0078	0.1378	11.32×	0.0034	0.0015	**0.0061**	0.1280	7.82×
	SRU	0.0259	0.0016	0.0079	0.0422	12.47×	0.0285	0.0016	0.0066	0.1048	12.79×	0.0280	0.0014	0.0079	0.1221	9.66×
	UltraRE	0.0279	0.0017	0.0084	0.0407	17.99×	0.0322	**0.0006**	**0.0048**	0.0977	16.22×	0.0246	0.0020	0.0083	0.1031	13.40×
	IFRU	0.0020	0.0011	0.0063	0.0020	**0.06**×	0.0017	0.0008	0.0055	0.0023	**0.07**×	0.0024	0.0012	0.0064	0.0024	**0.10**×
	SURF	**0.1595**†	**0.0008**	**0.0058**†	**0.6080**†	0.50×	0.1235	0.0011	0.0058	**0.5384**†	0.66×	0.0708	**0.0009**†	0.0071	0.4821	0.51×
ML-100k	Retrain	**0.0945**†	0.0043	0.0156	0.6017	1.00×	**0.0991**	0.0043	0.0156	**0.7514**†	1.00×	**0.0658**	**0.0018**†	0.0123	**0.5211**†	1.00×
	RecEraser	0.0452	0.0034	0.0190	0.1287	3.64×	0.0422	0.0058	0.0212	0.1264	4.18×	0.0328	0.0010	0.0122	0.1065	3.98×
	SRU	0.0490	0.0031	0.0201	0.1360	2.63×	0.0504	0.0046	0.0178	0.1641	1.70×	0.0464	0.0032	0.0178	0.0924	2.91×
	UltraRE	0.0471	**0.0028**	0.0201	0.1202	4.23×	0.0597	0.0040	0.0156	0.1802	4.83×	0.0362	0.0013	0.0100	0.1188	4.71×
	IFRU	0.0053	0.0058	0.0188	0.0060	**0.45**×	0.0095	0.0035	0.0178	0.0049	**0.42**×	0.0036	0.0031	0.0189	0.0055	**0.46**×
	SURF	0.0864	0.0041	**0.0145**†	**0.6653**†	0.58×	0.0871	**0.0027**†	**0.0156**	0.5580	0.69×	0.0490	0.0023	**0.0100**	0.4053	0.65×
Steam	Retrain	**0.0832**†	**0.0003**	**0.0017**	**0.4385**†	1.00×	**0.0424**†	0.0003	0.0018	**0.4468**†	1.00×	**0.0793**†	**0.0003**	0.0019	**0.6488**†	1.00×
	RecEraser	0.0537	0.0007	0.0020	0.0655	2.28×	0.0333	0.0005	0.0023	0.0670	2.25×	0.0293	0.0005	0.0022	0.0917	2.28×
	SRU	0.0571	0.0005	0.0022	0.0878	1.34×	0.0409	0.0004	0.0022	0.1137	1.31×	0.0421	0.0006	0.0023	0.1946	1.35×
	UltraRE	0.0614	0.0004	0.0019	0.0594	2.24×	0.0392	0.0004	0.0021	0.0674	2.39×	0.0325	0.0004	0.0024	0.1445	6.81×
	IFRU	0.0669	0.0006	0.0018	0.0188	**0.02**×	0.0008	0.0005	0.0020	0.0006	**0.02**×	0.0008	0.0004	0.0022	0.0004	**0.01**×
	SURF	0.0721	0.0004	0.0020	0.3319	0.04×	0.0189	**0.0002**	**0.0016**	0.1645	0.04×	0.0423	0.0003	0.0020	0.2542	0.04×
FS-TKY	Retrain	0.4199	0.0011	0.0051	**0.6566**†	1.00×	**0.3161**†	0.0007	**0.0014**†	**0.7801**†	1.00×	**0.2773**†	0.0006	0.0022	**0.6780**†	1.00×
	RecEraser	0.2452	0.0009	0.0028	0.1978	3.13×	0.1591	0.0008	0.0028	0.1804	3.00×	0.1654	0.0004	0.0027	0.2782	3.14×
	SRU	0.0912	0.0011	0.0023	0.2123	5.72×	0.0908	0.0015	0.0023	0.2563	5.17×	0.0905	0.0006	0.0032	0.2638	5.74×
	UltraRE	0.1420	0.0010	0.0036	0.1839	3.31×	0.1025	0.0009	0.0028	0.2193	3.11×	0.1223	0.0005	0.0022	0.1955	3.29×
	IFRU	0.1253	**0.0002**†	**0.0014**†	0.0341	**0.11**×	0.0008	0.0002	0.0019	0.0008	**0.09**×	0.0001	0.0001	**0.0004**†	0.0006	**0.11**×
	SURF	**0.4371**	0.0012	0.0028	0.5972	0.36×	0.2487	**0.0002**	0.0018	0.2598	0.35×	0.1238	**0.0001**	0.0014	0.1252	0.36×
FS-NYC	Retrain	0.3565	0.0011	0.0029	**0.6013**†	1.00×	**0.2892**	0.0002	0.0019	**0.6917**†	1.00×	**0.2115**†	0.0003	0.0048	**0.6132**†	1.00×
	RecEraser	0.1473	0.0017	0.0039	0.1612	2.87×	0.0979	0.0003	0.0020	0.1109	2.98×	0.0803	0.0003	0.0009	0.1593	2.91×
	SRU	0.0396	0.0001	0.0010	0.1424	3.80×	0.0375	**0.0001**	**0.0010**	0.1780	3.73×	0.0368	0.0002	0.0019	0.1698	2.68×
	UltraRE	0.1057	**0.0001**	0.0010	0.1002	4.75×	0.0904	0.0002	0.0010	0.0902	4.75×	0.0533	0.0009	0.0048	0.0742	4.78×
	IFRU	0.1218	0.0002	0.0018	0.0322	**0.25**×	0.0008	0.0001	0.0010	0.0006	**0.22**×	0.0008	**0.0002**	**0.0020**	0.0004	**0.26**×
	SURF	**0.3780**†	0.0003	**0.0019**	0.5004	0.57×	0.2575	0.0005	0.0029	0.3192	0.60×	0.1773	0.0004	0.0029	0.3321	0.59×
Yelp	Retrain	**0.0337**	0.0003	0.0006	0.4560	1.00×	**0.0227**†	0.0003	0.0004	**0.3860**†	1.00×	**0.0188**†	0.0001	0.0004	**0.5503**	1.00×
	RecEraser	0.0132	0.0002	0.0004	0.0458	2.22×	0.0103	**0.0001**	0.0003	0.0421	2.17×	0.0063	0.0001	0.0006	0.0165	2.20×
	SRU	0.0075	0.0001	**0.0003**	0.0175	1.30×	0.0046	0.0001	**0.0003**	0.0187	1.27×	0.0031	0.0002	0.0005	0.0080	1.30×
	UltraRE	0.0077	0.0001	0.0004	0.0157	1.18×	0.0051	0.0002	0.0005	0.0098	1.15×	0.0026	0.0002	0.0005	0.0089	1.17×
	IFRU	0.0121	0.0002	0.0005	0.0041	**0.01**×	0.0001	0.0003	0.0006	0.0001	**0.01**×	0.0003	0.0002	0.0006	0.0001	**0.01**×
	SURF	0.0304	**0.0001**	0.0004	**0.4867**†	0.04×	0.0145	0.0002	0.0004	0.2980	0.04×	0.0122	**0.0001**	**0.0004**	0.2513	0.04×

Hyperparameters. Increasing k markedly degrades recommendation quality while forgetting remains stable, since a smaller neighbourhood confines the subtraction: $k = 1$ gives the best trade-off, though larger values may suit unpopular items. The scaling factor α modulates the trade-off between utility and forgetting strength: low values suppress recommendations aggressively, yielding near-zero NDCG but strong unlearning, while higher values preserve quality at a marginal cost in UL MRR@20.

Empirical validation of the theory. The high FRBO values in Table 1 indicate that the ordering of relevant items is largely unchanged after unlearning, as predicted by the ranking-stability result. Measuring the score shift $|\tilde{y}(S_u^l) - f_{\mathcal{M}}(S_u^l)|$ against the embedding distance $\|\mathbf{e}_u - \mathbf{e}_{i^*}\|$ on ML-100k with SASRec, the perturbation stays limited and approximately constant inside $\mathcal{N}_k(\mathbf{e}_{i^*})$, with no sharp increase at its boundary and well below the worst-case bound of Eq. (2). Finally, the score difference $f_{\mathcal{M}}(S_u^l) - \tilde{y}(S_u^l)$ tracks the scaled auxiliary output $\alpha f_{\mathcal{M}'}(S_u^l)$ closely, and the residual $f_{i^*}(S_u^l) - f_{\mathcal{M}'}(S_u^l)$ concentrates around zero, so ε is small in practice; since that difference has a positive mean (≈ 0.83), the removed component contributes substantially to the original predictions and the operation is non-trivial.

5. Conclusions

We introduced SURF, a lightweight and model-agnostic framework for machine unlearning in SRSs that forgets an item by identifying its local embedding neighbourhood, training a compact auxiliary model on that region and subtracting its influence at inference time. On seven datasets and three architectures, SURF matches or surpasses existing baselines in unlearning metrics while preserving ranking quality, improving NDCG by up to 32% over the best baseline and requiring just 2% of the retraining time budget against the up to 17× overhead of shard-based methods. Because the original parameters remain untouched, unlearning requests can also be reverted instantly. Future work will extend SURF to other domains, such as image and text.

Declaration of Generative AI

In the preparation of this paper, generative AI tools, specifically ChatGPT and Claude, were used to assist with grammar checking, language refinement, and the improvement of clarity and coherence. These tools served as supplementary aids in the writing process. All AI-generated suggestions were reviewed, evaluated, and revised by the authors to ensure the accuracy, relevance, and appropriateness of the content. The authors take full responsibility for the final content of this paper, including its accuracy, originality, and academic integrity.

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